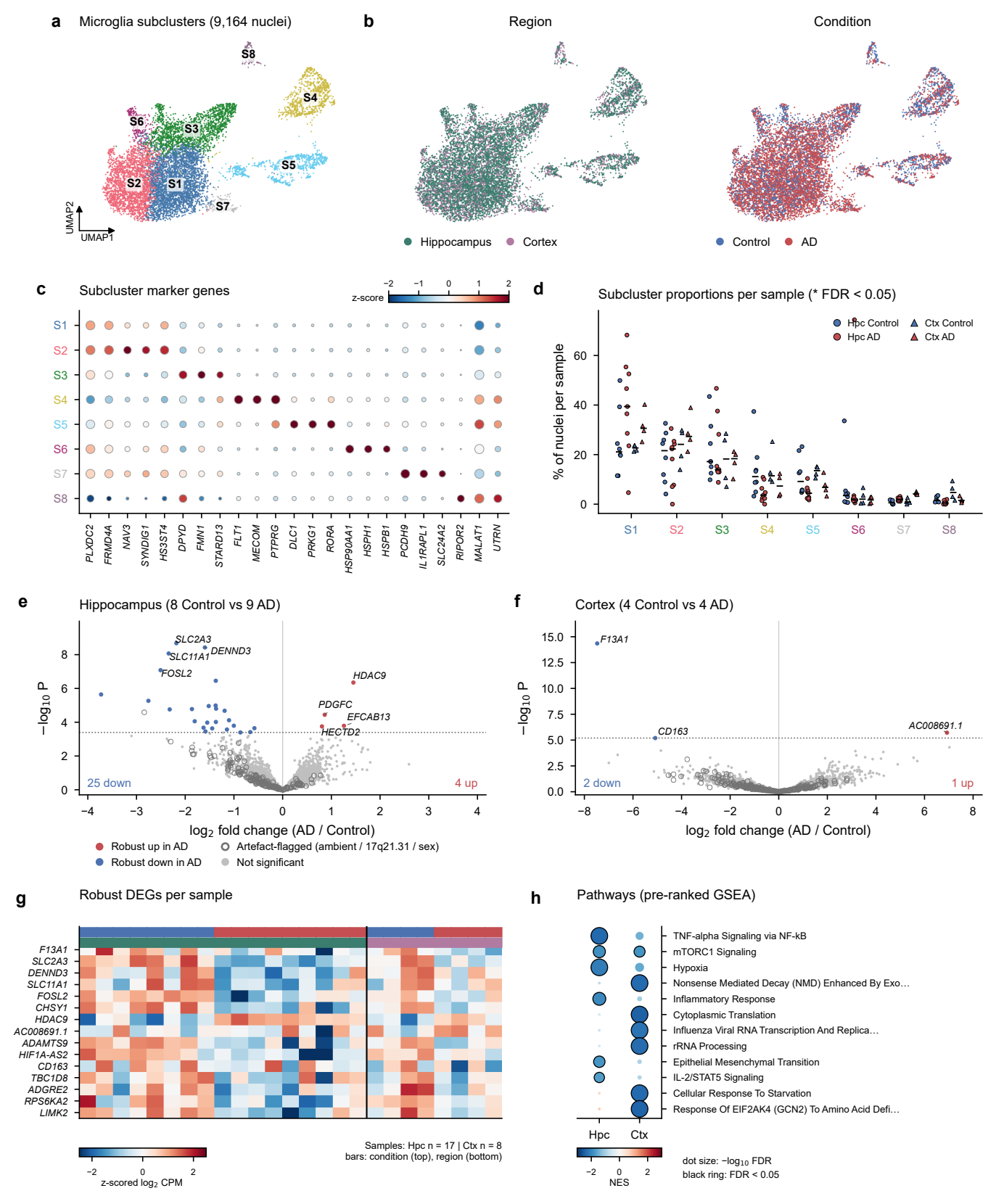

a
Microglia subclusters (9,164 nuclei)
S8
S4
S6
S3
S5
S2
S1
S7
UMAP2
UMAP1
b
Region
Hippocampus
Cortex
Condition
Control
AD
c
Subcluster marker genes
z-score
−2 −1 0 1 2
S1
S2
S3
S4
S5
S6
S7
S8
PLXDC2
FRMD4A
NAV3
SYNDIG1
HS3ST4
DPYD
FMN1
STARD13
FLT1
MECOM
PTPRG
DLC1
PRKG1
RORA
HSP90AA1
HSPH1
HSPB1
PCDH9
IL1RAPL1
SLC24A2
RIPOR2
MALAT1
UTRN
d
Subcluster proportions per sample (* FDR < 0.05)
Hpc Control
Hpc AD
Ctx Control
Ctx AD
% of nuclei per sample
0
20
40
60
S1
S2
S3
S4
S5
S6
S7
S8
e
Hippocampus (8 Control vs 9 AD)
SLC2A3
SLC11A1
DENND3
FOSL2
HDAC9
PDGFC
EFCAB13
HECTD2
−log10 P
25 down
4 up
log2 fold change (AD / Control)
Robust up in AD
Robust down in AD
Artefact-flagged (ambient / 17q21.31 / sex)
Not significant
f
Cortex (4 Control vs 4 AD)
F13A1
CD163
AC008691.1
−log10 P
2 down
1 up
log2 fold change (AD / Control)
g
Robust DEGs per sample
F13A1
SLC2A3
DENND3
SLC11A1
FOSL2
CHSY1
HDAC9
AC008691.1
ADAMTS9
HIF1A-AS2
CD163
TBC1D8
ADGRE2
RPS6KA2
LIMK2
z-scored log2 CPM
Samples: Hpc n = 17 | Ctx n = 8
bars: condition (top), region (bottom)
h
Pathways (pre-ranked GSEA)
TNF-alpha Signaling via NF-kB
mTORC1 Signaling
Hypoxia
Nonsense Mediated Decay (NMD) Enhanced By Exo…
Inflammatory Response
Cytoplasmic Translation
Influenza Viral RNA Transcription And Replica…
rRNA Processing
Epithelial Mesenchymal Transition
IL-2/STAT5 Signaling
Cellular Response To Starvation
Response Of EIF2AK4 (GCN2) To Amino Acid Defi…
Hpc
Ctx
NES
dot size: −log10 FDR
black ring: FDR < 0.05